Cover page (paste into the template cover). Keep the template's PJDSC 2026 header and theme line.

BANTAY-BAHA

A Graph-Based Early Warning and Response System for Flood-Linked Disease Outbreaks

Trace the Pattern, Target the Cure: Advancing Public Health Through Data Science

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Revised concept proposal (v3). It keeps the structure and limits of the submitted proposal and changes only what the build corrected or changed; Appendix D lists each change and why. Anonymous submission: no team or member names.

EXECUTIVE SUMMARY

249 words (limit 250)

Eight to nine tropical cyclones cross the Philippines each year, and flood-linked disease follows within weeks: Typhoon Ondoy (2009) was followed by 2,089 leptospirosis cases and 162 deaths, and the 2019 dengue epidemic reached 146,062 cases by July. Response still begins after hospitals fill, and the communities most at risk are often the ones a storm has cut off from care.

Bantay-Baha ("flood watch") is a decision-support dashboard that tells DOH regional offices and LGU health officers, before or immediately after a typhoon, which provinces are most likely to see a disease surge and which of those will lose access to a hospital. It links three graphs: (1) a flood-exposure layer from Sentinel-1 flood extent, UP NOAH hazard maps and river-routed rainfall; (2) a causal lead-lag outbreak graph learned from province-week DOH dengue surveillance with PCMCI+, which identifies sentinel provinces that rise first; and (3) a road-and-RoRo ferry network that simulates storm-closed routes to measure lost hospital access. The first two feed a quantile gradient-boosted forecast of 1- to 4-week-ahead incidence, benchmarked against time-series foundation models and scored as forecast hubs do. Calibrated surge probability and the population cut off from care combine into one priority ranking with recommended action tiers.

We backtest on Typhoons Ompong (2018), Tisoy (2019), Rolly and Ulysses (2020), plus every other 2017-2020 storm. No Philippine tool conditions outbreak risk on a specific storm and joins it with storm-specific loss of healthcare access; Bantay-Baha moves response from reacting to outbreaks to getting ahead of them.

KEYWORDS

early warning system, dengue, leptospirosis, flooding, typhoon, graph analytics, causal discovery, probabilistic forecasting, healthcare access, Philippines

BACKGROUND

294 words

About 20 tropical cyclones enter the Philippine Area of Responsibility each year and eight or nine cross the country, mostly between July and October (PAGASA, n.d.). Each storm brings flooding, and flooding brings disease. After Typhoon Ondoy in 2009, more than 2,089 people were treated for leptospirosis and 162 died, over five times the national leptospirosis deaths of 2008 (Balbuena et al., 2010); Manila hospital data show the leptospirosis risk from rainfall peaks two weeks later and is mediated by flooding (Matsushita et al., 2018). Dengue rises with the same rains: in 2019 the DOH declared a national epidemic after 146,062 cases and 622 deaths from January to July (The Philippine Star, 2019). Recent reviews call for a unified surveillance registry and tools that prioritise where to act (Interior et al., 2025; Labana, 2026).

The data needed to act early exist but are scattered across surveillance, hazard, climate and facility datasets. Dengue forecasting from climate has been studied repeatedly in the Philippines, from province-level early warning (Buczak et al., 2014) to Metro Manila machine learning (Carvajal et al., 2018) and a nationwide comparison of eight forecasters (Olana et al., 2025). Flood-driven loss of hospital access has been modelled for Metro Manila, where a 15 cm flood leaves about 22% of residents without access to higher-level care (Tariverdi et al., 2023), and dengue source regions have been identified by synchrony analysis (van Panhuis et al., 2015). These strands have never been joined: no tool conditions surge risk on

a specific typhoon, models nationwide loss of access including ferry links, or combines the two into one ranking for health officers. The problems share a structure that graph methods model directly: water flows along rivers, outbreaks move between connected places, and patients travel along roads and ferries.

THE PROBLEM

110 words

Philippine health authorities have no tool that anticipates, before or immediately after a typhoon, where flood-linked disease surges will occur and which affected communities will be cut off from care. Prophylaxis, vector control and medical teams are deployed after cases rise, and isolated communities are reached last. Existing dengue models are not storm-conditioned, give point forecasts without calibrated uncertainty and ignore whether patients can reach a hospital; hazard tools such as UP NOAH carry no health layer.

Guiding question: Can observed flood extent, a causal graph of outbreak spread and a storm-disrupted transport network, combined with calibrated machine-learning forecasts, rank the provinces that need help first, early enough to act?

OBJECTIVES

227 words

By the PJDSC 2026 final-round submission, the team will:

- By 24 October 2026, assemble a province-by-week panel for 85 harmonised province units (the dengue file's 126 province and city units, crosswalked), 2016 to 2020, joining DOH-Epi dengue cases, CHIRPS and IMERG rainfall, Sentinel-1 flood fraction, UP NOAH hazard share and IBTrACS typhoon exposure on PSGC codes.
- By 24 October 2026, learn a lead-lag outbreak graph with PCMCI+, identify sentinel provinces (PageRank) and communities (Louvain), and test whether sentinel features reduce follower-province forecast error out of sample.
- By 24 October 2026, forecast 1- to 4-week-ahead dengue incidence per province with a quantile gradient-boosted model, achieving a relative Weighted Interval Score below 1.0 against the seasonal-naive baseline and 90% interval coverage of 80 to 95% on rolling-origin evaluation over 2018 to 2019.
- By 24 October 2026, quantify storm-specific loss of hospital access for every city and municipality with a friction-surface and a road-plus-RoRo network method for Ompong, Tisoy, Rolly and Ulysses, reporting their disagreement as a range, with sensitivity bounds and the Metro Manila benchmark of Tariverdi et al. (2023).
- By 24 October 2026, deploy the dashboard that returns a ranked province list with calibrated risk, access loss, sentinel status, drivers and an action tier for any historical or shifted track in under one minute, and before the final-round submission collect System Usability Scale scores from three target users.

SCOPE & LIMITATIONS

213 words

Scope. Nationwide, the 85 province units that the 2016 to 2020 DOH-Epi dengue file on HDX supports, at weekly resolution; access analysis per city and municipality, summarised per province as the share of

population cut off. Dengue is the primary disease because consistent subnational data exist. Leptospirosis is a stretch goal: an open weekly series exists for 12 cities (Project CCHAIN), so it enters as a city and national lag analysis and a typhoon event study, not as a province forecast. Storm scenarios use historical IBTrACS tracks and shifted variants. Users are DOH regional offices, LGU health offices and local DRRM offices.

Limitations. Surveillance data carry reporting delays and under-reporting, and COVID-19 collapsed 2020 reporting, so 2020 is used for access scenarios only. Sentinel-1 misses urban and vegetated floods and can miss the peak; tracks without an image use hazard maps, which give an upper bound. Hazard maps cover 74 of 85 units, and a unit with no map is missing, not zero hazard. OpenStreetMap completeness outside Luzon is uneven, so two access methods are reported; hospital lists and 2024 to 2026 ferry routes are proxies. PCMCI+ links are statistical dependencies, not proof of transmission. The tool supports decisions and does not replace clinical or field judgment; it uses only aggregated, non-identifiable data.

METHODOLOGY

754 words

Bantay-Baha builds three graphs, turns each into features for one province-week table, and combines a calibrated disease forecast with storm-driven access loss into one priority ranking (Figure 1, Appendix A; data sources in Table 1, Appendix B).

Layer 1: Flood exposure (prevent)

For each typhoon, Sentinel-1 radar before-and-after change detection in Google Earth Engine (UN-SPIDER practice, same-orbit images, a backscatter-drop rule) yields the flooded fraction and flooded population (WorldPop 100 m) per province and municipality, cross-checked against Copernicus EMS observed flood (EMSR312, Ompong; EMSR478 grades damage only). Static susceptibility is the share of population inside UP NOAH 5- and 25-year flood hazard maps. Dynamic forcing uses CHIRPS and GPM IMERG 7-day rainfall and rainfall accumulated downstream along the HydroRIVERS network, so towns near river mouths register upstream rain.

Layer 2: Causal lead-lag outbreak graph (stop outbreaks early)

Province-week incidence is log-transformed and seasonally differenced so shared rainy-season peaks do not create false links. PCMCI+ (Runge, 2020) in Tigramite, with partial-correlation tests and lags up to three weeks, recovers a directed lead-lag graph with bootstrap confidence per edge and Benjamini-Hochberg control of false links; lagged cross-correlation is the baseline graph. PageRank ranks sentinel provinces that rise first; Louvain groups provinces that move together for coordinated response. Graphs are learned only inside each training window to avoid leakage.

Forecast model

A pooled XGBoost model with province as a categorical feature predicts 1- to 4-week-ahead incidence from lagged cases, calendar week, lagged rainfall, flood exposure (Layer 1) and lagged cases of PCMCI+ leader provinces (Layer 2), trained with the multi-quantile objective at the 23 levels used by the CDC FluSight hub and explained with SHAP. Models without graph or environmental features are the ablations. Baselines set the bar: zero-shot Chronos-2 (Ansari et al., 2025) and TabPFN-TS, a seasonal-naive forecast and an MPNN-LSTM on province-border and ferry adjacency; their median ensemble with XGBoost is evaluated too. Because only XGBoost takes the storm track, storm scenarios use XGBoost. Surge probability is recalibrated by isotonic regression on past forecasts only (Zadrozny & Elkan, 2002); an online conformal band checks the intervals (Gibbs & Candès, 2021). Models train on earlier years and are evaluated by

rolling-origin re-forecasts, never a random split, scored with the Weighted Interval Score (Bracher et al., 2021), relative WIS and coverage.

Layer 3: Access graph and storm scenarios (strengthen healthcare reach)

Nodes are OSM road junctions and hospitals from OSM and UP NOAH facilities (the DOH facility registry blocks scripted download). Edges are OSM road links and the 108 RoRo routes in the MARINA inventory (roroph), weighted by travel time and frequency. Population on a 1 km grid joins the nearest road, and a multi-source shortest-path search gives each cell's travel time to the nearest hospital. For each typhoon, ferry routes crossing the IBTrACS gale swath are removed and road links inside the observed flood are cut; shifted tracks, which have no image, use UP NOAH hazard inside the swath. Cells that lose their route, or whose trip doubles and grows by over an hour, are cut off; Meta HRSL counts the under-5 and 60+ among them. A raster method, the Malaria Atlas Project friction surface (Weiss et al., 2020) with flooded cells impassable, runs in parallel following Petricola et al. (2022); the disagreement between methods is reported as uncertainty.

Priority score and product

For a chosen typhoon, each province's priority is P = R x A, where R is the calibrated probability that incidence over the next four weeks exceeds the province's 90th percentile and A is the share of population cut off from hospital care, so P reads as the expected share of residents at elevated risk who cannot reach care. The Streamlit dashboard takes a historical or shifted track and returns a priority map, a risk-versus-access 2x2, a ranked table with predicted cases and intervals, cut-off population, sentinel status and SHAP drivers, and an action tier: monitor sentinel sites; pre-position doxycycline and vector-control supplies; or deploy mobile clinics before roads and ferries close. Scenarios are pre-computed (15 tracks per backtest typhoon, plus every other 2017-2020 storm), so results return in under one minute.

Validation

We backtest the forecast on Ompong (2018), Tisoy (2019) and the 2019 season, and the access module on all four typhoons, checking whether high-priority provinces saw higher incidence in the following four weeks. Sensitivity to travel speed, ferry rules and storm buffer is reported as rank stability (Kendall's tau) for both access methods. Three target users complete a 15-minute scenario task and the System Usability Scale.

Tools

Python with pandas, GeoPandas, GDAL, SciPy and scikit-image (road graph, shortest paths and cost-distance), Tigramite, XGBoost, scikit-learn, SHAP, statsmodels, chronos-forecasting, tabpfn-time-series and PyTorch; Google Earth Engine; Streamlit; all code in a public GitHub repository with a one-command rebuild.

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APPENDICES

Appendix A. Figure 1. Bantay-Baha pipeline: three graphs, two models, one ranking

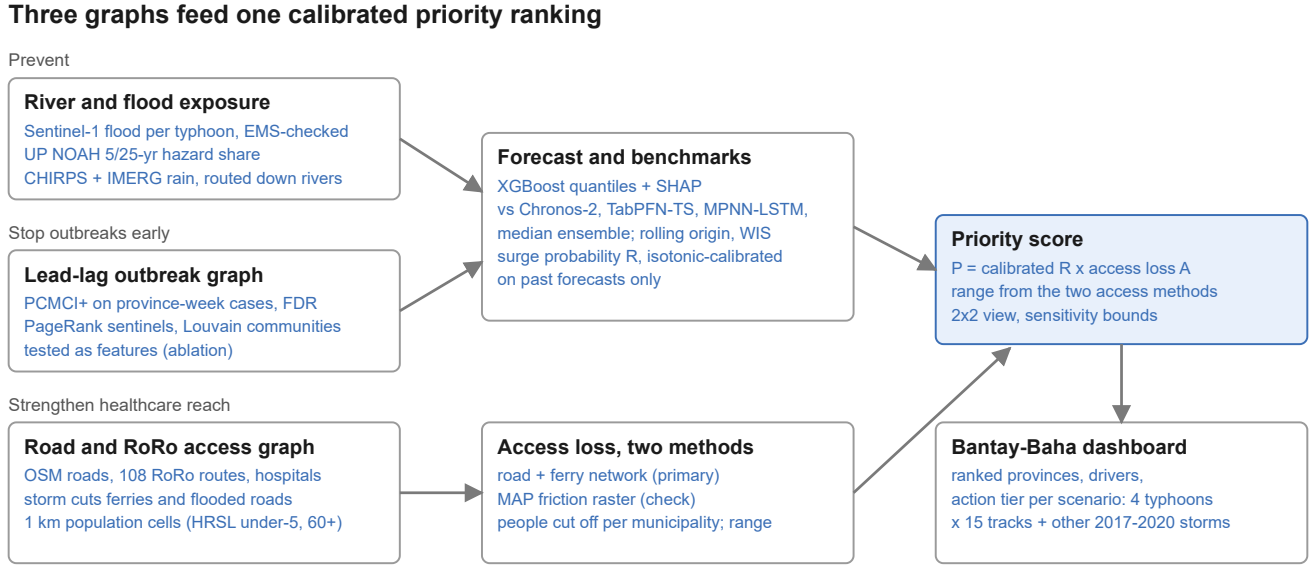


Figure 1. The flood and lead-lag graphs become features of a quantile forecast; the access graph becomes a population-cut-off estimate from two methods; the two meet only in the priority score, so each can be validated on its own.

Appendix B. Table 1. Data sources

Data	Source	Role
Dengue cases, 126 province and city units x week, 2016-Jan 2021 (deaths not used)	DOH Epidemiology Bureau via Humanitarian Data Exchange (DOH-Epi Dengue Data 2016-2021)	Forecast target; lead-lag graph
National weekly dengue, leptospirosis and other notifiable diseases, 2017-2023	Mendeley Data (Parreño, 2024; v98325dpxv); OpenDengue V1.3	Lepto national lag analysis; cross-check
City weekly leptospirosis and dengue, 12 cities	Project CCHAIN PIDSR city-week (HDX)	Lepto city lag analysis; typhoon event study
Flood extent per typhoon	Sentinel-1 SAR change detection in Google Earth Engine (after the UN-SPIDER practice); Copernicus EMS EMSR312 (observed flood), EMSR478 (damage grading only); Global Flood Database (no Philippine event after 2017; context only)	Observed flood exposure
Flood hazard maps, 5/25-year	UP NOAH (DOST) per-province hazard maps	Static susceptibility; flood proxy for tracks without an image
Daily rainfall	GPM IMERG V07 (0.1°); CHIRPS v2.0 (0.05°)	Dynamic forcing
River network with downstream links	HydroRIVERS v1.0 Australasia (the Asia file has no Philippine rivers)	Upstream rainfall routing

Typhoon tracks	IBTrACS v04r01 (basin WP)	Storm scenarios; storm catalogue
Roads and ferry routes	OpenStreetMap (Geofabrik Philippines); MARINA RoRo inventory via roroph (108 routes, 2024-2026 operations)	Access graph edges
Hospital locations	HOT OSM health facilities (HDX); UP NOAH critical facilities; DOH National Health Facility Registry (optional, manual)	Access graph destinations
Friction surface and travel time	Malaria Atlas Project 2019 (Weiss et al., 2020)	Raster access method
Population and vulnerability	WorldPop 2020 100 m; Meta High Resolution Population Density (under-5, 60+)	Population weights; vulnerable people cut off
Boundaries, PSGC codes, census	HDX COD-AB (PSA/NAMRIA); 2020 CPH; reviewed crosswalk to 85 units	Join keys

Appendix C. Table 2. Development plan, 13 October to final-round submission (revised)

Dates	Workstream	Deliverable	Status
13-22 Oct	Data acquisition and checks	Manifest of verified source files; data findings; crosswalk to 85 units	done
23-24 Oct	Province-week panel; flood and climate layers (GEE)	Parquet panel and data card; Sentinel-1 flood, IMERG, NOAH shares	done
23-24 Oct	Forecasting and lead-lag graph	PCMCI+ graph, quantile XGBoost, calibrated R, WIS table	done
24 Oct	Benchmarks	Chronos-2, TabPFN-TS, MPNN-LSTM, median ensemble on the same rows	done
23-24 Oct	Access network	Network and friction travel times, cut-off population per scenario, sensitivity, storm catalogue	done
24 Oct	Dashboard and integration	Streamlit app with pre-computed scenarios and a built-in usability test	done
to submission	User test, video, deck, repository	SUS with three target users; 3-minute demo; 20 slides; README with one-command rebuild	in progress

Appendix D. Changes from the submitted proposal

#	Submitted proposal	Revised	Why
1	Panel for all 81 provinces	85 harmonised 2016-2020 province units	The DOH file has 126 reporting units that mix provinces with cities and use pre-2022 boundaries; a reviewed crosswalk folds cities into provinces, undoes the Maguindanao split and the Special Geographic Area, and keeps Metro Manila as four districts.
2	Dengue cases and deaths	Cases only	2016 deaths imply a case fatality several times that of other years.
3	LiPAD 5-, 25- and 100-year hazard maps	UP NOAH 5- and 25-year maps	The per-province maps come from the UP NOAH archive; 11 units have no 25-year map and are carried as missing. The 25-year map is the road-flood proxy, so the 100-year map is not used.
4	Sentinel-1 cross-checked against EMSR312 and EMSR478	EMSR312 only	EMSR478 (Rolly) has damage grading only, no flood delineation.
5	Sentinel-1 change detection (UN-SPIDER practice)	Same-orbit before/after, backscatter-drop rule	The after/before ratio on dB values with mixed orbits recovered almost none of the EMS-confirmed flood.
6	Global Flood Database as a flood-extent source	Context only	Its Philippine events end in 2017; it covers none of the four backtest storms.
7	HydroRIVERS v1.0 Asia	HydroRIVERS v1.0 Australasia	HydroSHEDS places the Philippines in the Australasia region; the Asia file has no Philippine rivers.
8	IMERG rainfall	CHIRPS and IMERG	Both were already in the data table; CHIRPS also drives river routing.
9	PCMCI+ graph	PCMCI+ with Benjamini-Hochberg control	Without it, PCMCI+ returned many false links on pure-noise checks.
10	"The final forecast is the median ensemble"	Ensemble evaluated; storm scenarios use XGBoost	The zero-shot models take no storm-track input, so they cannot condition on a shifted track; the ensemble has the best mean WIS, XGBoost the best median WIS.
11	Chronos-2, TabPFN-TS, MPNN-LSTM as specified	Chronos-2 on the incidence scale; TabPFN v2 weights; border + ferry adjacency	Chronos-2 on log incidence, as first specified, collapsed on one-week reporting spikes (reported too); the default TabPFN-TS weights need a login and licence; no fixed adjacency was given.
12	Conformal (EnbPI) intervals with MAPIE	Online conformal check from past errors	It fits the rolling origin without refitting the model (Gibbs & Candès, 2021).
13	Calibrated risk (no method named)	Isotonic recalibration of R on past forecasts only	The model's surge probability was under-confident in the middle range.
14	Leptospirosis: "no open subnational series exists"	City and national lag analysis, typhoon event study	Project CCHAIN publishes weekly city counts for 12 cities; nothing is extrapolated to provinces.
15	Hospitals from the DOH NHFR and OSM	HOT OSM + UP NOAH facilities	The NHFR site blocks scripted download; it stays an optional manual input.

16	Access nodes at city and municipality centres	1 km population cells joined to the road graph	Centres misplace spread-out populations; results are still summed per municipality.
17	Roads cut inside the observed flood polygon	Observed flood for real tracks; NOAH 25-year hazard for shifted tracks	A track that did not happen has no image; the proxy gives an upper bound, radar a lower bound.
18	Two access methods	Same, with the per-unit range shown as uncertainty	The methods agree on normal-day rankings but not on the size of a storm's cut.
19	Meta HRSL for population and vulnerability	Used for under-5 and 60+ counts cut off	HRSL under-counts people aged 60+ against the 2020 census; counts are low estimates.
20	Four backtest typhoons	Plus the observed track of every other 2017-2020 storm whose gale swath reaches land	Shows the tool on any historical track; storms of 2016 are skipped because their model would have under a year of history.
21	OSMnx in the tools	GDAL, SciPy and scikit-image	The nationwide graph is built from the Geofabrik extract and solved with compiled shortest-path and cost-distance routines.
22	Development plan, 13-24 October	Revised dates (Table 2)	The panel and model deadlines of 15 and 20 October were missed; the build ran on 23-24 October.